

EC602 — Computer Network

Semester VI • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Application Layer Protocols

Application Layer (Layer 7): Provides network services directly to user applications. Protocols define the format and rules for communication between applications on different hosts.

Protocol	Port	Transport	Function
DNS	53	UDP/TCP	Resolves domain names to IP addresses.
SMTP	25	TCP	Sends email from client to server (or server to server).
SNMP	161	UDP	Manages and monitors network devices.
FTP	20/21	TCP	File transfer; port 21 = control, port 20 = data.
HTTP	80	TCP	Transfers web pages (HyperText Transfer Protocol).
HTTPS	443	TCP	Secure HTTP using TLS/SSL encryption.

1.1 DNS (Domain Name System)

DNS

Function: Translates human-readable domain names (e.g. `www.google.com`) to IP addresses.

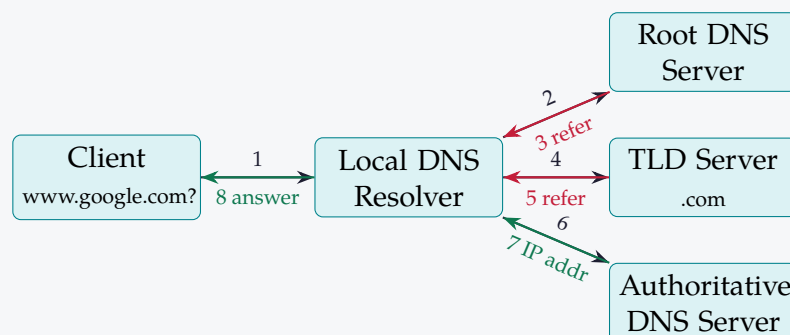
Hierarchy: Root DNS → TLD servers (.com, .org, .in) → Authoritative DNS servers.

Resolution types:

- **Recursive:** DNS resolver queries on behalf of client until answer found.
- **Iterative:** DNS server returns best referral; client queries next server itself.

Caching: DNS responses cached with TTL to reduce lookup time.

DNS Resolution Hierarchy



1.2 HTTP and WWW

HTTP (HyperText Transfer Protocol): Application layer protocol for transferring web content. Stateless, request-response model. Client sends HTTP request; server responds with status code and content.

HTTP Methods: GET (retrieve), POST (submit data), PUT (update), DELETE (remove), HEAD (headers only).

Common Status Codes: 200 OK, 301 Moved Permanently, 404 Not Found, 500 Internal Server Error.

WWW: World Wide Web — a system of interlinked hypertext documents accessed via HTTP. Uses URLs, HTML, and browsers.

2. Network Security

2.1 Cryptography

Cryptography: The science of securing information by transforming it into an unreadable format (ciphertext) using an algorithm and a key. Only authorised parties with the correct key can decrypt it.

Symmetric Key (Private Key) Cryptography

Symmetric Key Cryptography

Concept: Same key used for both encryption and decryption. Both sender and receiver must share the secret key securely.

Algorithms: DES (56-bit key), 3DES, AES (128/192/256-bit), RC4, Blowfish.

Advantages: Fast, low computational overhead, suitable for bulk data encryption.

Disadvantages: Key distribution problem — how to securely share the key? Does not scale well (n users need $n(n-1)/2$ keys).

Asymmetric Key (Public Key) Cryptography

Asymmetric Key Cryptography

Concept: Uses a **key pair** — a **public key** (shared openly) and a **private key** (kept secret). Data encrypted with public key can only be decrypted with the corresponding private key.

Algorithms: RSA, Diffie-Hellman, ECC (Elliptic Curve Cryptography).

Encryption: Sender encrypts with receiver's **public key**; receiver decrypts with **private key**.

Advantages: Solves key distribution problem; enables digital signatures.

Disadvantages: Computationally expensive; much slower than symmetric encryption.

Feature	Symmetric	Asymmetric
Keys used	Same key (shared secret)	Public + Private key pair
Speed	Fast	Slow
Key distribution	Difficult (must share secretly)	Easy (public key is open)
Scalability	Poor ($n^2/2$ keys)	Good ($2n$ keys for n users)
Examples	AES, DES, RC4	RSA, Diffie-Hellman, ECC
Use case	Bulk data encryption	Key exchange, digital signatures

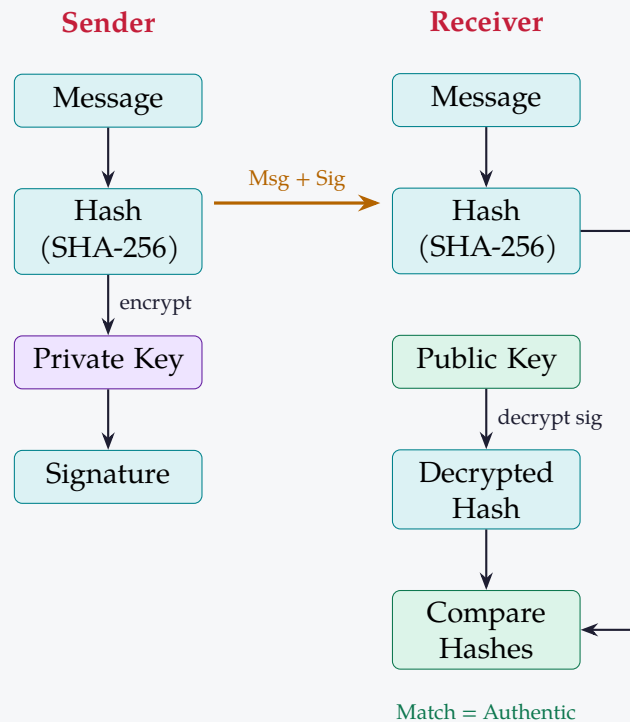
2.2 Digital Signature

Digital Signature: A cryptographic mechanism that provides **authentication**, **integrity**, and **non-repudiation**. Created by encrypting a message digest (hash) with the sender's **private key**.

Process:

1. Sender computes hash of message using a hash function (e.g. SHA-256).
2. Sender encrypts the hash with their **private key** — this is the digital signature.
3. Receiver decrypts the signature using sender's **public key** to get the hash.
4. Receiver independently computes hash of received message.
5. If both hashes match: message is authentic and unaltered.

Digital Signature — Sign & Verify



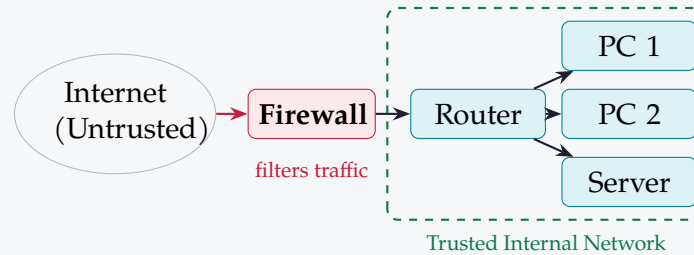
2.3 Firewalls

Firewall: A network security device (hardware or software) that monitors and controls incoming/outgoing network traffic based on predefined security rules. Acts as a barrier between trusted internal network and untrusted external network.

Types of Firewalls:

Type	Description
Packet Filter	Inspects packets at network layer; filters based on IP, port, protocol. Stateless.
Stateful Inspection	Tracks connection state; allows only packets belonging to established connections.
Application Gateway	Proxy-based; inspects application layer data (HTTP, FTP). Most secure but slowest.
Circuit-level Gateway	Monitors TCP handshake; does not inspect packet content.

Firewall Placement in a Network



3. Modern Topics

3.1 ISDN (Integrated Services Digital Network)

ISDN: A set of communication standards for simultaneous digital transmission of voice, video, data, and other services over traditional telephone copper wire. Replaced by DSL and broadband.

ISDN Channels:

- **B channel (Bearer):** 64 Kbps; carries user data (voice, video, data).
- **D channel (Delta):** 16 or 64 Kbps; carries signalling and control information.

ISDN Interfaces:

Interface	Description
BRI (Basic Rate)	$2B + D = 2 \times 64 \text{ Kbps} + 16 \text{ Kbps D} = 144 \text{ Kbps}$. For home/small office.
PRI (Primary Rate)	$23B + D$ (North America) or $30B + D$ (Europe) = 1.544 / 2.048 Mbps. For enterprises.

3.2 ATM (Asynchronous Transfer Mode)

ATM: A cell-switching and multiplexing technology that uses fixed-size **53-byte cells** (5-byte header + 48-byte payload) for transmitting voice, video, and data over high-speed networks.

Key Features:

- Fixed 53-byte cells enable predictable, low-latency switching.
- Connection-oriented: virtual circuits (VC) established before data transfer.
- Supports QoS guarantees for different traffic types (CBR, VBR, ABR, UBR).
- **VPI/VCI:** Virtual Path Identifier / Virtual Channel Identifier used for routing cells.

3.3 DSL (Digital Subscriber Line)

DSL: A family of technologies that transmit digital data over existing telephone copper wire pairs at high speeds. Uses frequencies above the voice band (above 4 kHz).

Type	Speed	Description
ADSL (Asymmetric)	Down: 8 Mbps, Up: 1 Mbps	Different download/upload speeds. Most common for home use.
SDSL (Symmetric)	2 Mbps both ways	Equal upload/download. Used by businesses.
VDSL (Very High)	Down: 52 Mbps, Up: 16 Mbps	Very high speed over short distances.

DSL Modem: Connects user's equipment to the telephone line. Separates voice (0–4 kHz) and data (above 4 kHz) using a **splitter**.

3.4 Cable Modem

Cable Modem: A device that provides Internet access over the cable TV (CATV) infrastructure using coaxial cable. Uses **DOCSIS** (Data Over Cable Service Interface Specification) standard.

Key Points:

- Shared medium: all users in a neighbourhood share the same cable segment (bandwidth shared).
- Downstream (TV to user): 6 MHz channel, up to 30–40 Mbps.
- Upstream (user to TV headend): narrower band, typically 5–42 MHz.
- **HFC (Hybrid Fibre-Coaxial):** Fibre from headend to neighbourhood node; coaxial to homes.

3.5 Wireless LAN — IEEE 802.11

IEEE 802.11 (Wi-Fi): A set of standards for wireless local area networking. Uses radio frequencies (2.4 GHz and 5 GHz bands). MAC protocol: CSMA/CA.

IEEE 802.11 Standards:

Standard	Year	Frequency	Max Speed
802.11b	1999	2.4 GHz	11 Mbps
802.11a	1999	5 GHz	54 Mbps
802.11g	2003	2.4 GHz	54 Mbps
802.11n	2009	2.4/5 GHz	600 Mbps (MIMO)
802.11ac	2013	5 GHz	1+ Gbps

Architecture:

- **Infrastructure mode:** Devices connect through an **Access Point (AP)**. AP connects to wired network. Most common.
- **Ad-hoc mode:** Devices communicate directly without an AP (peer-to-peer).
- **BSS (Basic Service Set):** Single AP + associated stations.
- **ESS (Extended Service Set):** Multiple BSSs connected via a distribution system.

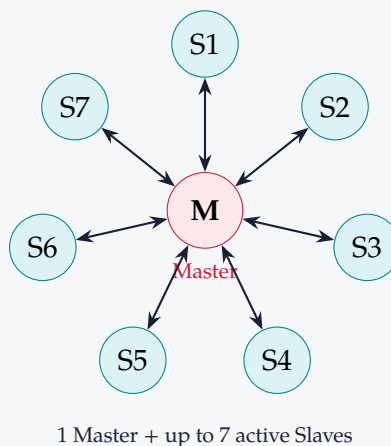
Security: WEP (weak, deprecated) → WPA → WPA2 (AES-based, current standard) → WPA3.

3.6 Bluetooth

Bluetooth (IEEE 802.15.1): A short-range wireless technology for personal area networks (PAN). Operates at 2.4 GHz ISM band. Range: ~10 m (Class 2). Uses **FHSS** (Frequency Hopping Spread Spectrum) — hops 1600 times/second across 79 channels.

Bluetooth Network (Piconet):

- **Piconet:** One **master** + up to 7 active **slaves**. Master controls timing and frequency hopping.
- **Scatternet:** Multiple overlapping piconets; a device can be slave in multiple piconets or master in one and slave in another.

Bluetooth Piconet Structure**Bluetooth Versions:**

Version	Key Feature
1.2	Basic rate 1 Mbps; AFH (Adaptive Frequency Hopping)
2.0 + EDR	Enhanced Data Rate; up to 3 Mbps
4.0 (BLE)	Bluetooth Low Energy; very low power; IoT devices
5.0	2× speed, 4× range, 8× broadcast capacity vs. 4.2

Quick Revision — Module 4

Application Layer, Security & Modern Topics | EC602

Key Facts at a Glance

DNS port	53 (UDP/TCP)
SMTP port	25 (TCP)
FTP ports	21 (control), 20 (data)
HTTP / HTTPS ports	80 / 443
SNMP port	161 (UDP)
Symmetric key	Same key for encrypt/decrypt (AES, DES)
Asymmetric key	Public key encrypts; private key decrypts (RSA)
Digital signature	Hash encrypted with sender's private key
ISDN BRI	$2B + D = 144$ Kbps
ISDN PRI (Europe)	$30B + D = 2.048$ Mbps
ATM cell size	53 bytes (5 header + 48 payload)
ADSL speed	Down 8 Mbps / Up 1 Mbps
802.11b speed	11 Mbps @ 2.4 GHz
802.11g/a speed	54 Mbps
Bluetooth range	~10 m; 2.4 GHz; FHSS; 1600 hops/sec
Piconet	1 master + up to 7 active slaves

One-Line Definitions

- **DNS:** Translates domain names to IP addresses using a hierarchical distributed database.
- **HTTP:** Stateless request-response protocol for transferring web content.
- **SMTP:** Protocol for sending email between mail servers.
- **FTP:** Protocol for transferring files; uses separate control (21) and data (20) connections.
- **Symmetric Cryptography:** Same secret key used for both encryption and decryption.
- **Asymmetric Cryptography:** Public key encrypts; private key decrypts (or vice versa for signatures).
- **Digital Signature:** Hash of message encrypted with sender's private key; proves au-

thenticity.

- **Firewall:** Security device that filters network traffic based on rules.
- **ISDN:** Digital telephone network supporting voice, video, and data simultaneously.
- **ATM:** Cell-switching technology using fixed 53-byte cells; supports QoS.
- **DSL:** High-speed Internet over telephone copper wire using frequencies above voice band.
- **IEEE 802.11:** Wi-Fi standard; uses CSMA/CA; operates at 2.4/5 GHz.
- **Bluetooth:** Short-range PAN technology; 2.4 GHz; FHSS; piconet architecture.

Frequently Asked / Viva Questions

Q1. What is DNS and how does it work?

DNS translates domain names to IP addresses. Client queries local DNS resolver; if not cached, resolver queries root → TLD → authoritative DNS servers recursively or iteratively.

Q2. What is the difference between symmetric and asymmetric cryptography?

Symmetric: same key for encrypt/decrypt, fast, key distribution problem. Asymmetric: public/private key pair, slower, solves key distribution.

Q3. How does a digital signature work?

Sender hashes the message and encrypts the hash with their private key. Receiver decrypts with sender's public key and compares hashes to verify authenticity and integrity.

Q4. What is a firewall?

A security device that monitors and filters network traffic based on predefined rules, acting as a barrier between trusted and untrusted networks.

Q5. What is the difference between packet filter and stateful firewall?

Packet filter: stateless, inspects each packet independently. Stateful: tracks connection state, allows only packets belonging to established connections.

Q6. What is ATM and what is its cell size?

Asynchronous Transfer Mode; cell-switching technology using fixed 53-byte cells (5-byte header + 48-byte payload).

Q7. What is ADSL?

Asymmetric DSL; provides higher download speed (up to 8 Mbps) than upload speed (up to 1 Mbps) over telephone copper wire.

Q8. What MAC protocol does IEEE 802.11 use?

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

Q9. What is a piconet in Bluetooth?

A small network of one master and up to 7 active slave devices communicating on the same frequency hopping sequence.

Q10. What is FHSS in Bluetooth?

Frequency Hopping Spread Spectrum; Bluetooth hops across 79 channels 1600 times per second to reduce interference and improve security.

Q11. What is the difference between infrastructure and ad-hoc Wi-Fi mode?

Infrastructure: devices connect through an Access Point. Ad-hoc: devices communicate directly without an AP.

Q12. What is ISDN BRI?

Basic Rate Interface: $2B + D$ channels = $2 \times 64 \text{ Kbps} + 16 \text{ Kbps} = 144 \text{ Kbps}$ total. Used for home/small office ISDN connections.

Application Layer Protocols Summary

Protocol	Port	Transport	Purpose
DNS	53	UDP/TCP	Name resolution
SMTP	25	TCP	Send email
FTP	20/21	TCP	File transfer
HTTP	80	TCP	Web pages
HTTPS	443	TCP	Secure web
SNMP	161	UDP	Network management